

Stable Arm-Swing Humanoid Locomotion under Upper-Body Perturbations

A stage reward-routing curriculum preserves full-body waling, introduces coordinated shoulder-pitch arm swing, and evaluates recovery under torso, shoulder, and elbow action pulses.

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Track 2 Course Project: Training the upper body control through HumanoidVerse

? Problem formulation

Once the upper body is unlocked, humanoid walking is no longer just a lower-body tracking problem. Motions from the torso, shoulders, and elbows change the distribution of angular momentum, contact timing, and posture stability. If an arm-swing reward is added directly on top of the locomotion reward, the policy may learn unstable arm motions, excessive torso compensation, or sacrifice forward speed. We therefore frame the course objective as a stability problem: how to release upper-body degrees of freedom while preserving forward locomotion, and how to verify that the policy can recover from random upper-body perturbations.

★ Contributions

1. Full-body H1 control in a 19DoF action space.
2. Command-routed reward families that protect walking before arm shaping.
3. Upper-body action-pulse evaluation on torso, shoulder, and elbow channels.
4. Qualitative temporal evidence paired with finite-horizon metrics.
5. arm-swing reward can degrade the base gait.

💡 Design Principle

Locomotion is protected before expressiveness is added. Command derived masks activate standing, walking, and straight walk reward terms only in compatible regimes.

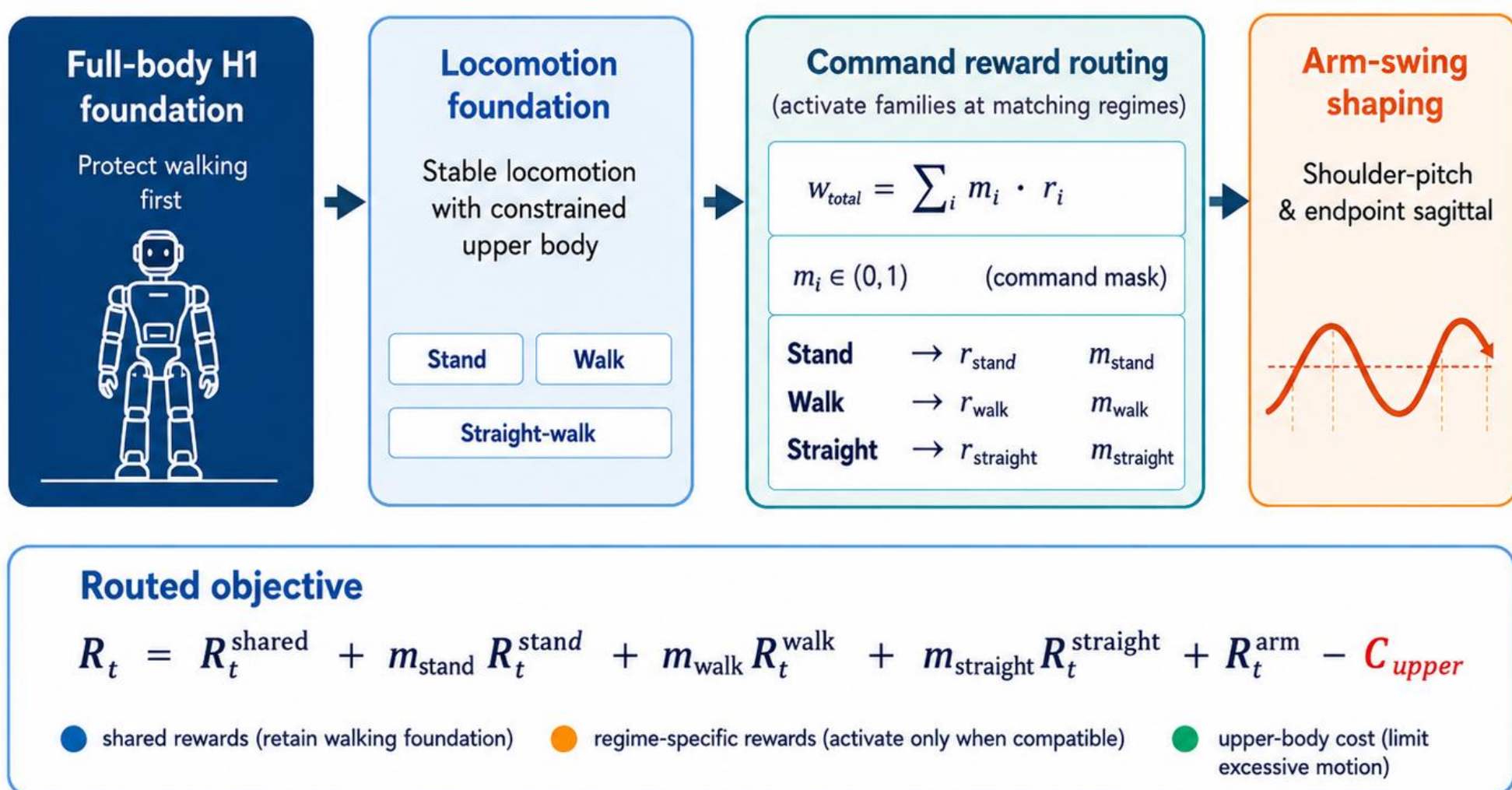
Dynamic Evidence: from Locked Upper Body to Perturbation Recovery



Evaluation Protocol

We evaluate 32 parallel H1 humanoids for a 20s horizon. Upper-body action pulses are applied to torso, shoulder, and elbow channels while the forward command remains active in a straight walk task.

Method: Staged Reward Routing



The staged protocol preserves the locomotion solution before strengthening upper-body expressiveness. Robustness is improved separately in the perturbation results, not mixed into the reward routing profiles.

The controller is trained in the full 19DoF H1 action space. Stable locomotion is protected first; command masks then activate stand, walk, and straight-walk reward families so arm-swing objectives appear mainly when they are compatible with forward walking.

Quantitative Robustness under Upper-Body Pulses



100%

survival at amp = 2.0 across 32 environments



0 vs. 95

falls: robust policy vs. arm-swing baseline under amp2

POLICY / SETTING	AMP	SURVIVAL	MEAN TIME	FALLS	TRACK ERR. (m/s)	MAX TILT (°)
Robust policy	0.0	100%	20.00s	0	0.0228	4.67°
Robust policy	1.0	100%	20.00s	0	0.0225	7.81°
Robust policy	2.0	100%	20.00s	0	0.0245	9.12°
Arm-swing baseline	2.0	0%	5.48s	95	0.1401	89.93°

Robust policy under increasing perturbation amplitude

	AMP 0.0	AMP 1.0	AMP 2.0
Survival	100%	100%	100%
Tracking error (m/s)	0.0228	0.0225	0.0245
Max tilt (°)	4.67°	7.81°	9.12°

The amp0 run uses the same perturbation scheduling with zero magnitude. At amp2, the robust policy retains arm motion: shoulder-pitch swing amplitude 0.2536 rad and elbow sagittal separation 0.1165 m, so robustness is not obtained by simply freezing the upper body.

Limitations

- Evidence is simulation only and short-horizon.
- Upper body naturalness is shaped by hand designed phase and amplitude rewards rather than human motion priors.
- Future work should add angular momentum analysis, energy cost, longer horizon, and cleaner camera evaluation.

Conclusion

Simply unlocking more joints is not enough to produce stable upper-body motion. We first preserve locomotion in the full-body action space, then add command routed arm coordination, and finally test whether the gait remains stable under upper-body perturbations. The resulting policy walks forward stably, shows clear arm swing, and recovers from torso, shoulder, and elbow action pulses.

Experimental Design

